



JunctionGuard AI

ible Accident-Prone Junction Safety Intelligence System

*reactive accident records to proactive, explainable junction safety
ence."*

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TEAM NAME

Traffic Titans

MEMBERS

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THEME

Transportation & Logistics

PROBLEM STATEMENT

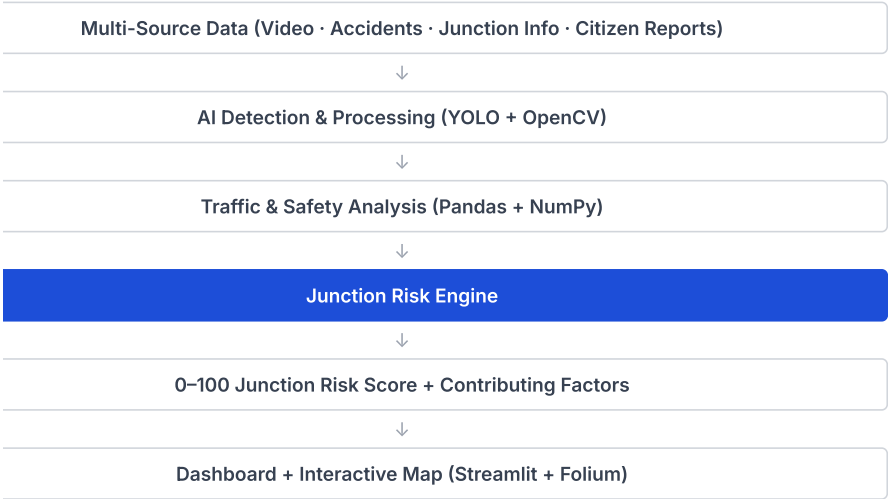
OMNI_TRANSPORT_12 (Improving Safety at Accident-Prone
Junctions)

POSED SOLUTION

02

eted Safety Intelligence Fusing Sensors, Data, and Citizen Science

Guard AI is a junction-level safety intelligence platform that fuses traffic-video observations, accident records, junction data and citizen reports into on
able Junction Risk Score.



l: Accident → Record → Analyze → React **JunctionGuard: Detect → Analyze → Score → Explain → Prevent**

WHAT MAKES IT DIFFERENT

- 1

Multi-Source Intelligence
Traffic feeds, historical data, and crowdsourcing combined into a dashboard.
- 2

Explainable Risk Scoring
Shows exactly which safety factors (speed, conflicts, pedestrians) contribute to each risk rating.
- 3

Geospatial Decision Support
Interactive Leaflet map displaying pulsing beacons to help author allocate patrol units.

SION. BUILD THE IMPOSSIBLE.

Junction

STEM ARCHITECTURE

Layered Data Engineering and Inference Pipelines

URCES	Traffic Video		Accident Records		Junction Information		Citizen Reports					
ON & SING	YOLOv8 Object Detection				OpenCV Frames Extraction							
ALYSIS	Pandas Dataframes				NumPy Vectorization							
INE LAYERS	Traffic Density		Speed Proximity Conflict		Pedestrian Activity		Accident History		Crowdsourced			
SE SYNC	Local SQLite Caching				Supabase Cloud Sync (PostgreSQL)							
ZATION	Python Scripts		Streamlit Dashboard GUI				Folium Map Layering					
DATA	DETECTION		ANALYSIS		RISK ENGINE		SCORE		EXPLANATION		DASHBOARD	

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Feasibility & Viability

04

Engineering Tech Feasibility, System Vulnerabilities, and Countermeasures

Technical Feasibility	Challenges & Risks	Mitigation Strategies
YOLOv8: Vehicle and pedestrian bounding box detection. OpenCV: Frame handling & video parsing. Pandas & NumPy: Accelerated statistical computations. PostgreSQL/SQLite: Real-time relational database logging. Leaflet & Folium: Highly responsive GIS interface.	■ Lighting Variations: Rainy conditions and night feeds reduce model visibility. ■ Missing Remote Data: Off-grid intersections lack telemetry connections. ■ Edge Compute Lag: YOLO pipelines lag during server deployments.	■ Preprocessed Filters: Adaptive thresholding to handle lighting changes. ■ Hybrid Sync: Local database caches (SQLite) acting as hot-fallbacks. ■ Zero-Latency Cache: Pre-computing telemetry indicators to eliminate processing lag.

FACT & BENEFITS

Creation Across Public, Operational, and Institutional Horizons

05

TRAFFIC AUTHORITIES	ROAD SAFETY TEAMS	URBAN PLANNERS	JUNCTION MGMT TEAM
SOCIAL BENEFITS Earlier identification of risky junction conditions to save lives. Better prioritization of traffic officer safety deployments. Improved pedestrian safety awareness. Active citizen participation via crowdsourced reporting.	ECONOMIC / OPERATIONAL ■ Optimal resource allocation for road repairs. ■ Data-driven infrastructure planning (signals, mirrors, speed bumps). ■ Granular risk metrics for insurance and government auditing.	LONG-TERM VALUE ■ Scalability to monitor hundreds of urban junctions. ■ Longitudinal accident trend profiling. ■ Integration with future Smart City sensor hubs.	

SEARCH & REFERENCES

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ic and Technological Groundwork Supporting the System

Object Detection

Johnson et al. "You Only Look Once: Unified, Real-Time Object Detection" (YOLO v1 work).

Geospatial Visualization

Folium Leaflet Engine - python-visualization.github.io/folium for map tiles and vector polygons.

Computer Vision Libraries

OpenCV (Open Source Computer Vision Library) documentation for hardware streaming.

Ministry Safety Norms

iRAP (International Road Assessment Programme) Star Rating methodology for accident-prone blackspots.

Safety Research

Global Status Report on Road Safety - guidelines for mitigating mixed urban environments.

Cloud Synchronization

Supabase client schemas for postgres CRUD replication and secure binary storage bucket upload.

Science Foundations

NumPy array matrix calculations & Pandas data processing pipeline manuals.

Database Architectures

SQLite SQLite3 modules - relational indexing and local transaction caching schema.

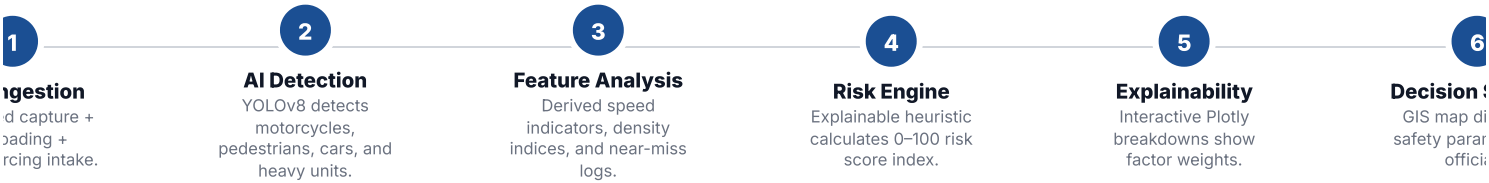
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THODOLOGY

age Implementation Pipeline Turning Raw Junction Signals into Preventive Action

07



OBSERVE → ANALYZE → SCORE → EXPLAIN → ACT

KEY FEATURES

08

Comprehensive Tools Built to Empower Highway Authorities & Traffic Marshals

1 AI Traffic Detection

YOLO-based classification of mixed vehicles and pedestrian units in real-time.

2

Traffic Analysis

OpenCV + NumPy metrics generating spatial conflict indexes and density maps.

3 Junction Risk Score

Interpretable 0–100 score dynamically calculated for each road junction.

4

Explainable Risk

Interactive factor weight diagrams indicating exact road vulnerabilities.

5 Crowdsourced Sentinel

Citizen report portal saving location, severity, and photo logs.

6

Interactive Safety Map

Folium mapping layer containing hazard buffers, heatmap overlays, and pulse beacons.

6 Analytics Dashboard

Streamlit UI for traffic monitoring, camera filtering, and logs audits.

8

Safety Recommendations

Auto-derived engineering solutions based on specific risk factors.

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JunctionGuard AI

PLAINABLE RISK SCORE

table Junction Ratings and Factor Contribution Matrices

LEVEL CLASSIFICATION



BUTING FACTORS

Vehicle Interaction / Conflict	Accident History
Pedestrian Activity	Traffic Density
Speed Indicator	Crowdsourced Reports



**Demonstrative calibration showing a simulated danger score base mixed conflict rates.*

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P & INNOVATION

1C

ishing Minimum Viable Scope from Long-Term Architectural Roadmap

MUST HAVE (MVP)	SHOULD HAVE (STANDARD)	BONUS / FUTURE (INNOVATION)
LO-based traffic detection. cident database analytical parsing. 100 Danger indexing calculation. plainable factor representation. eractive Leaflet mapping interface.	■ Citizen crowdsourced portal. ■ Media upload (images/videos) sync. ■ Time-series trend auditing. ■ Nearest-intersection lookup logic.	■ Automated civic dispatch ticket route ■ Helmets & triple-riding vision checker ■ Edge camera processor compilation. ■ Preemptive weather-linked risk forecasting.